Mikhail Vorobev

Latest version of this CV

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Germany English (B2), Russian (native)

Experience

JetBrains, Language Designer / Researcher 2025 – present

Research, design, prototype, and evolve Kotlin language features, investigating ecosystem needs and implementation constraints in coordination with compiler, libraries, and other teams.

- Build a K2 compiler-plugin prototype to detect potential data races in Kotlin coroutines with interprocedural May-Happen-in-Parallel static analysis.
- Design `lateinit val` / assign-once properties, including runtime semantics, the compilation scheme, and compatibility with frameworks.
- Prototype refinement types in the Kotlin compiler.
- Develop proposals to expose internal type-inference controls to library authors, including the Explicit Type Parameters KEEP.
- Investigate integrating virtual threads with Kotlin coroutines, focusing on performance trade-offs; co-author the KEEP research write-up.

Cloudless Labs (formerly Fluence Labs), Compiler Engineer 2023 – 2024

Designed Aqua, a language for orchestrating distributed computations across the Fluence peer-to-peer network, and developed its compiler toolchain in Scala.

- Achieved a **4x speedup in linking** on large projects through compiler optimization.
- Worked on parsing, type checking, IR passes, optimizations, and code generation.
- Researched and implemented language features, including Unison-inspired class-like constructs called abilities and support for folding over CRDT streams.
- Developed and maintained the Aqua extension for Visual Studio Code.

Mosaic Research, Backend Engineer 2022 – 2023

Developed and maintained C++ infrastructure for high-frequency cryptocurrency trading.

- Implemented an inter-service communication protocol using Aeron.
- Built CMake CI pipelines with builds, tests, caching, and automated dependency updates.
- Developed Python 3 scripts and supporting microservices.

Tinkoff, Backend Engineer 2021 – 2022

Developed and maintained the Scala microservices powering an online credit broker within the Credit and Brokerage Systems team.

- Integrated a facial recognition service into a fully automated lending workflow, enabling customers to apply for a loan by uploading their passport.
- Contributed to the development of an internal webhook service and its partner integrations.
- Integrated RabbitMQ and Kafka into backend services for message-based communication.
- Implemented MongoDB and PostgreSQL integrations for persistent application data.

Technical skills

Languages: Kotlin, Lean, Scala, C, C++, Python 3, TypeScript

Tools: Git, Docker, Gradle, GitHub Actions

Education

Constructor University Bremen 2024 – present

Bachelor's in Software, Data and Technology
JetBrains Scholarship holder.

Moscow Institute of Physics and Technology 2019 – 2022

Bachelor's (incomplete) in Engineering and Computer Science
Phystech School of Applied Mathematics and Informatics.

Courses

Constructor University Bremen - Semantics of Programming Languages 2026
Formalized a lambda calculus with subtyping in Lean and proved progress and preservation.

Constructor University Bremen - Compilers 2026
Implemented a compiler for the LaMa educational language targeting x86.

Moscow Institute of Physics and Technology - Concurrency 2021
Implemented a C++ concurrency library with fibers, futures, and a thread pool.

Moscow Institute of Physics and Technology - Distributed systems 2021
Implemented distributed consensus and replication algorithms in C++: Paxos, Multi-Paxos, and Raft.

Air Engineering School CanSat in Russia - Sounding rocket payload 2017 – 2019
Developed onboard firmware and peripheral drivers in C for AVR and STM32 microcontrollers.

Projects

Oolong | **Scala 3** 2022
Developed a library for compile-time query generation for document stores using Scala 3 macros, inspired by Quill.

librscs | **C, AVR** 2017 – 2019
Developed an Iridium 9602 modem driver in a peripheral-driver library for AVR microcontrollers.

Teaching

Algorithm and Code Training School (ACTS) - Visiting lecturer Astana, Kazakhstan
Delivered a lecture for school students on parsing, ASTs, name resolution, type inference, type checking, control-flow graphs, and static analysis. 2026

Summer School "Slon" - Volunteer tutor Pushchino, Russia
Taught school students formal languages, mathematical logic, functional programming in Scala, and introductory Python. 2021 – 2022